

The History of Computers

Prepared by:
R.V. Misa





What are computers?

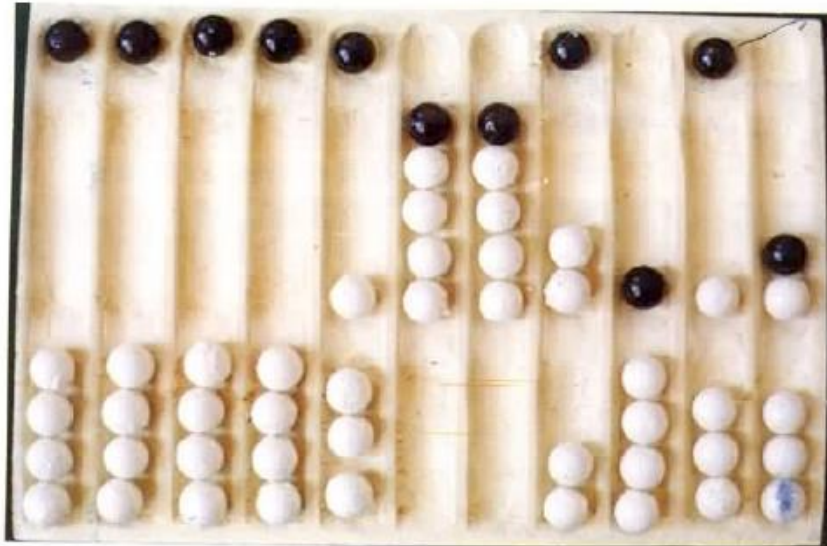
- The first computers were people! That is, electronic computers (and the earlier mechanical computers) were given this name because they performed the work that had previously been assigned to people. "Computer" was originally a job title: it was used to describe those human beings (predominantly women) whose job it was to perform the repetitive calculations required to compute such things as navigational tables, tide charts, and planetary positions for astronomical almanacs.



What are computers?

- Imagine you had a job where hour after hour, day after day, you were to do nothing but compute multiplications. Boredom would quickly set in, leading to carelessness, leading to mistakes. And even on your best days you wouldn't be producing answers very fast. Therefore, inventors have been searching for hundreds of years for a way to mechanize (that is, find a mechanism that can perform) this task.

Abacus



- The ***abacus*** was an early aid for mathematical computations. Its only value is that it aids the memory of the human performing the calculation. A skilled abacus operator can work on addition and subtraction problems at the speed of a person equipped with a hand calculator (multiplication and division are slower).



Abacus

- The abacus is often wrongly attributed to China. In fact, the oldest surviving abacus was used in 300 B.C. by the Babylonians. The abacus is still in use today, principally in the far east.

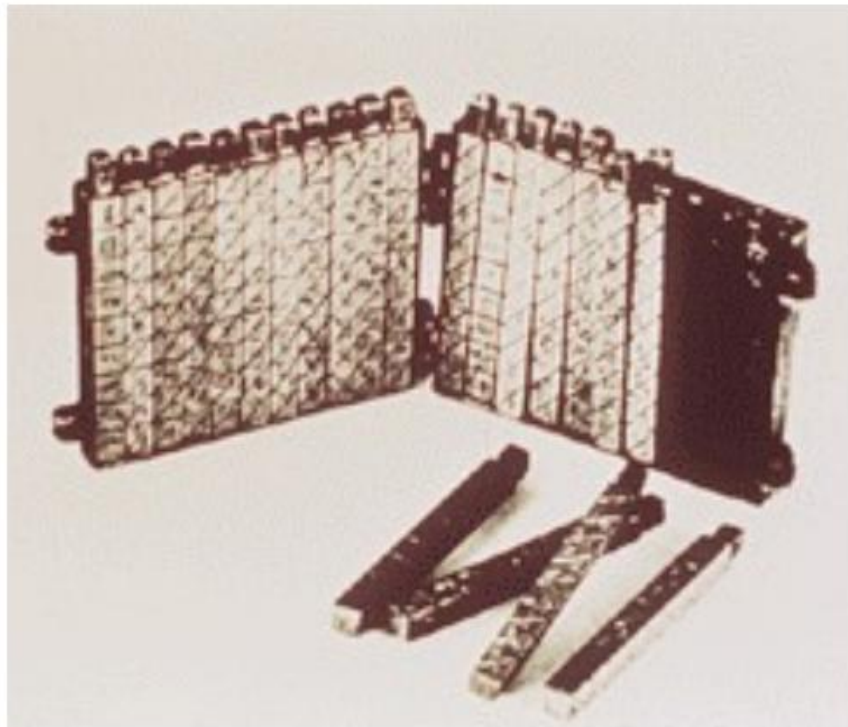




John Napier

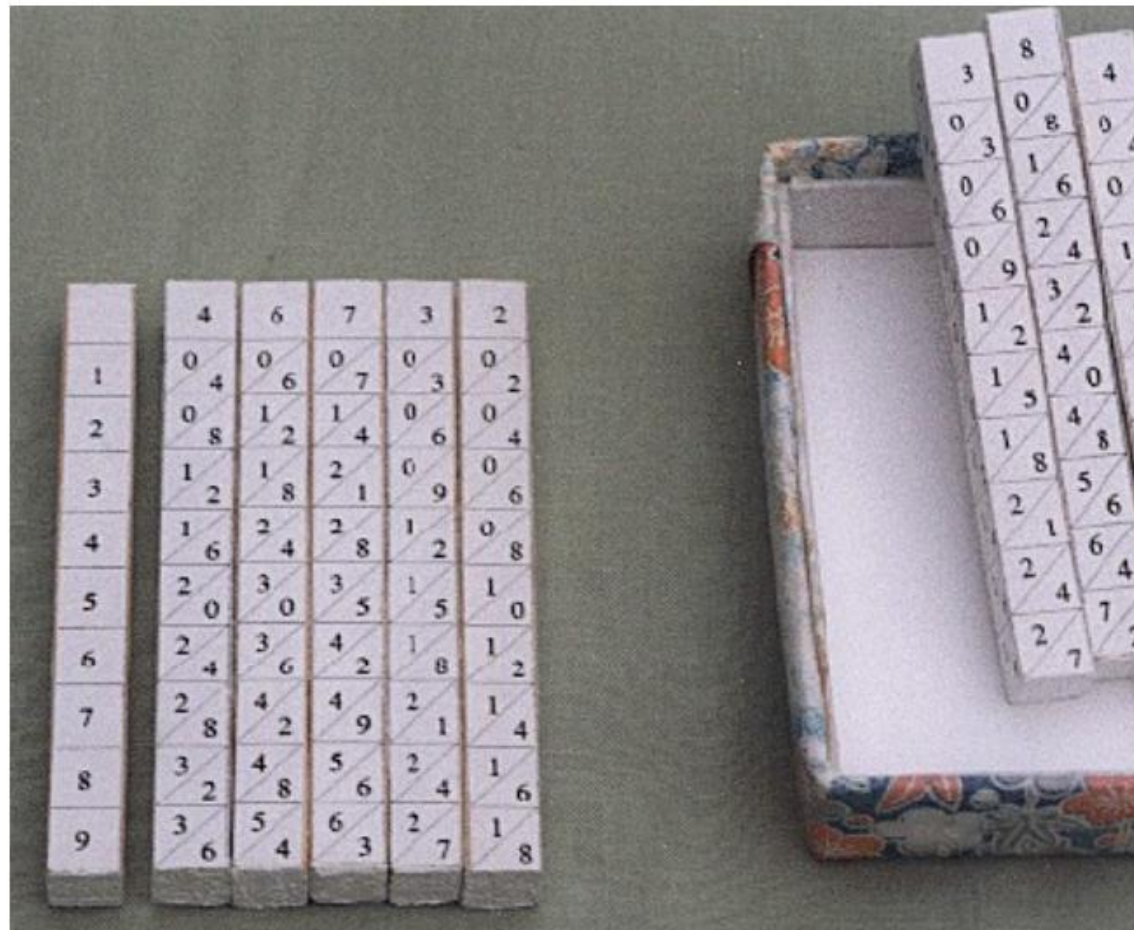
- In 1617 an eccentric (some say mad) Scotsman named John Napier invented ***logarithms***, which are a technology that allows multiplication to be performed via addition.
- Ex: $\log_2 x = 5$

Napier's Bones

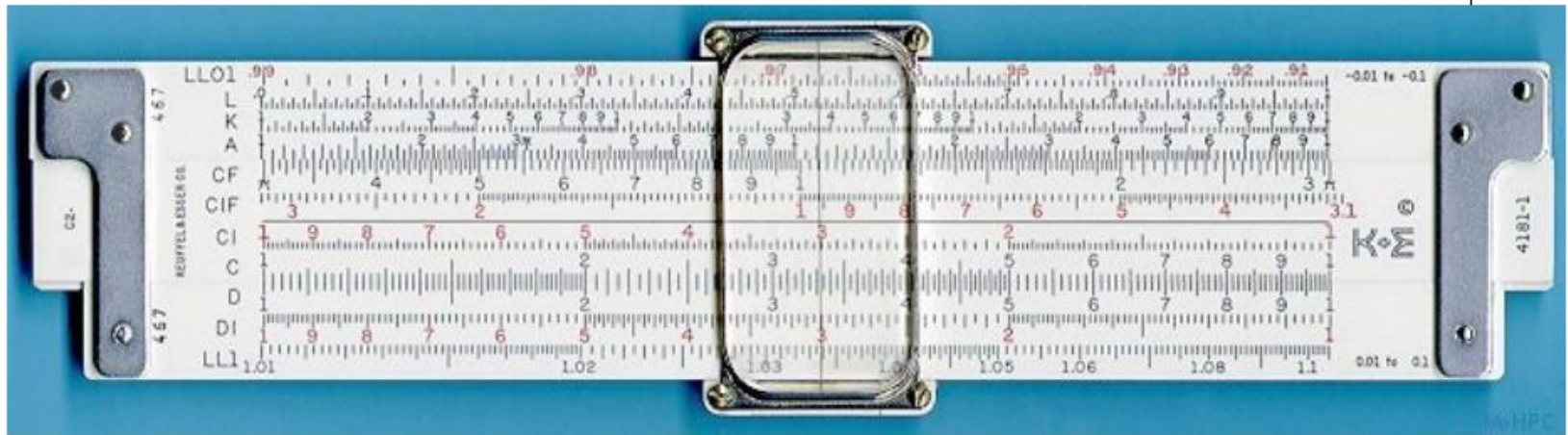


- The magic ingredient is the logarithm of each operand, which was originally obtained from a printed table. But Napier also invented an alternative to tables, where the logarithm values were carved on ivory sticks which are now called ***Napier's Bones***.

Napier's Bones



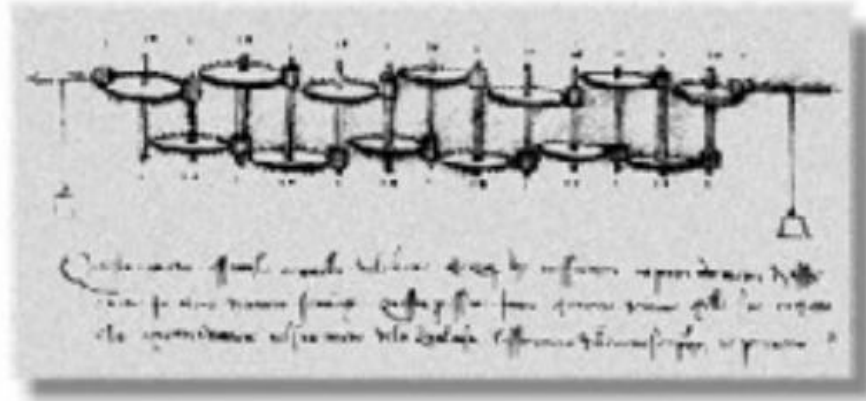
Slide Rule



- Napier's invention led directly to the **slide rule**, first built in England in 1632 and still in use in the 1960's by the NASA engineers of the Mercury, Gemini, and Apollo programs which landed men on the moon.

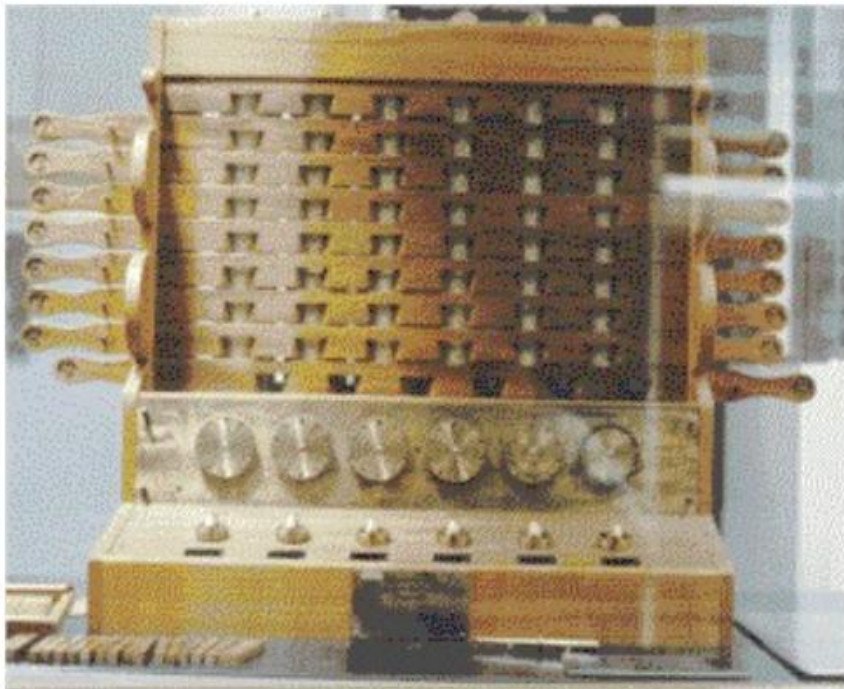


Leonardo da Vinci



- Leonardo da Vinci (1452-1519) made drawings of gear-driven calculating machines but apparently never built any.

Calculating Clock



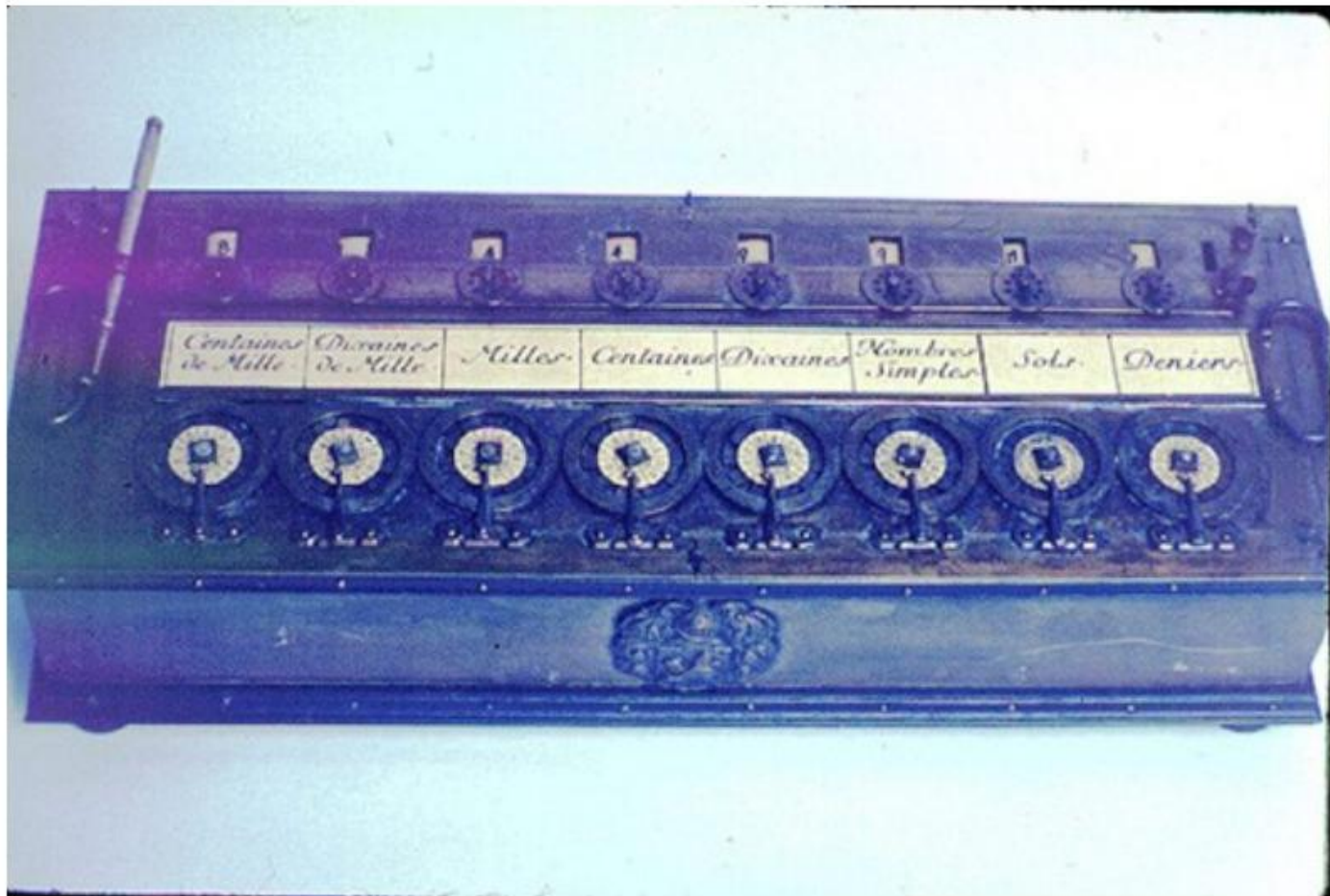
- The first gear-driven calculating machine to actually be built was probably the ***calculating clock***, so named by its inventor, the German professor Wilhelm Schickard in 1623. This device got little publicity because Schickard died soon afterward in the bubonic plague.



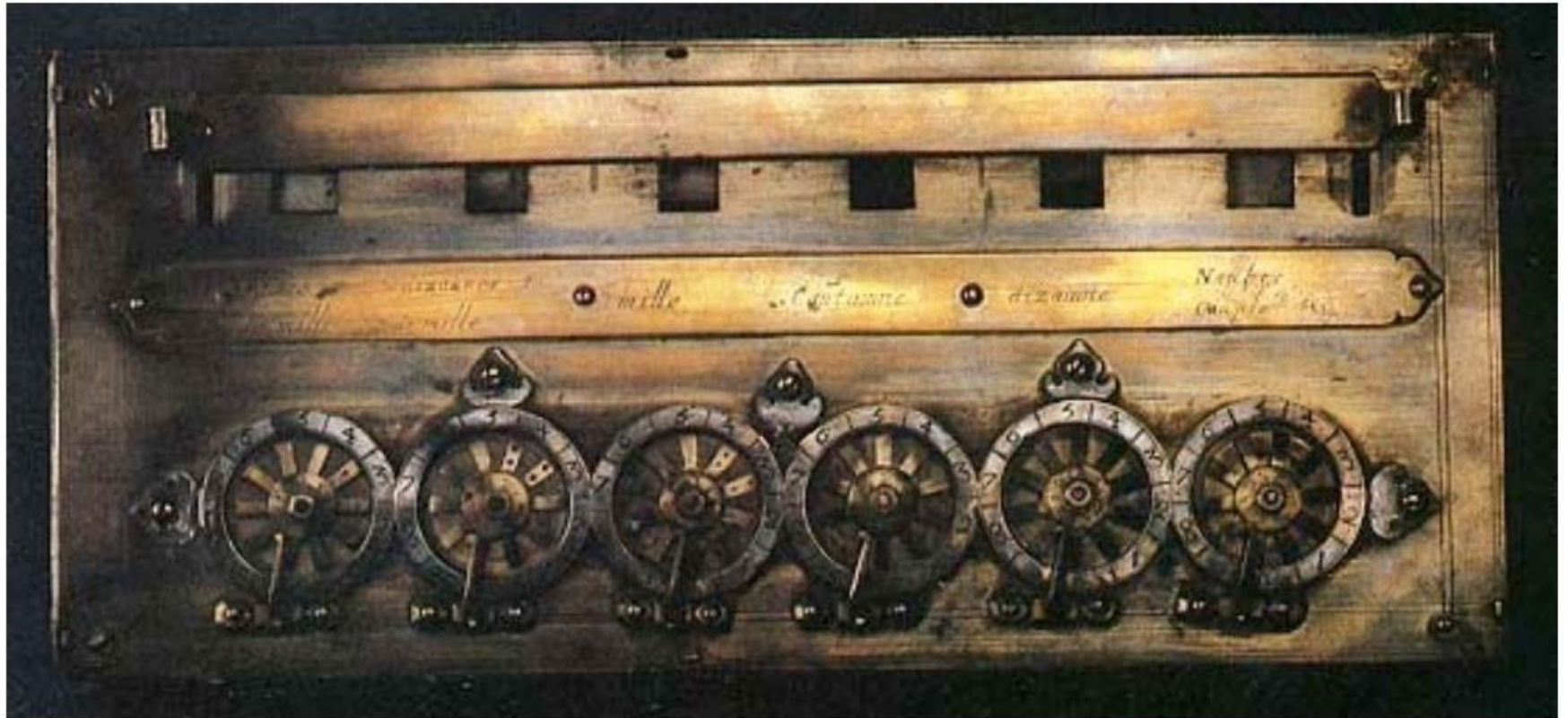
Blaise Pascal

- In 1642 Blaise Pascal, at age 19, invented the ***Pascaline*** as an aid for his father who was a tax collector. Pascal built 50 of this gear-driven one-function calculator (it could only add) but couldn't sell many because of their exorbitant cost and because they really weren't that accurate (at that time it was not possible to fabricate gears with the required precision).
- Up until the present age when car dashboards went digital, the odometer portion of a car's speedometer used the very same mechanism as the Pascaline to increment the next wheel after each full revolution of the prior wheel.

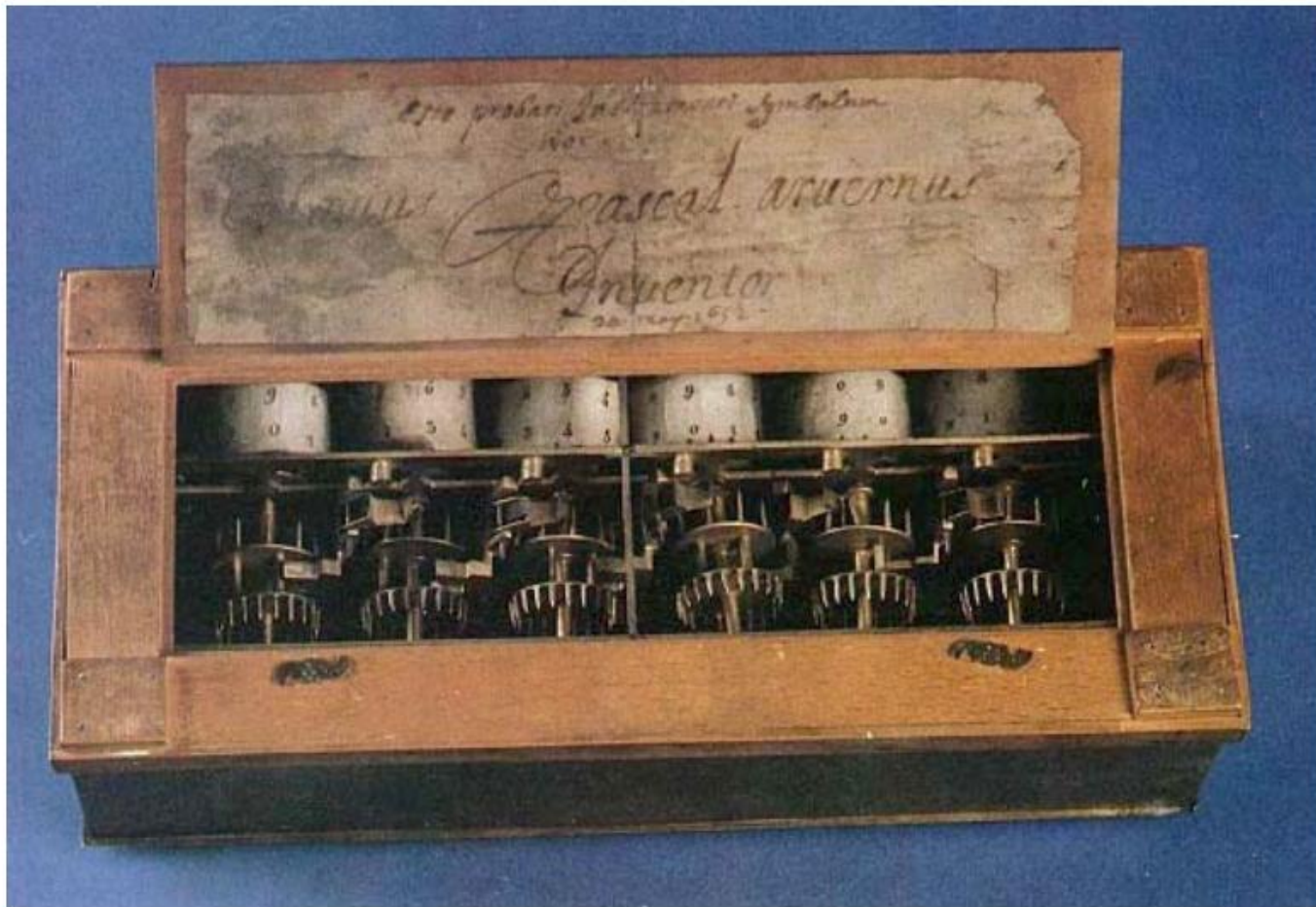
8-digit Pascaline



6-digit Pascaline (Cheaper)



Pascaline Insides

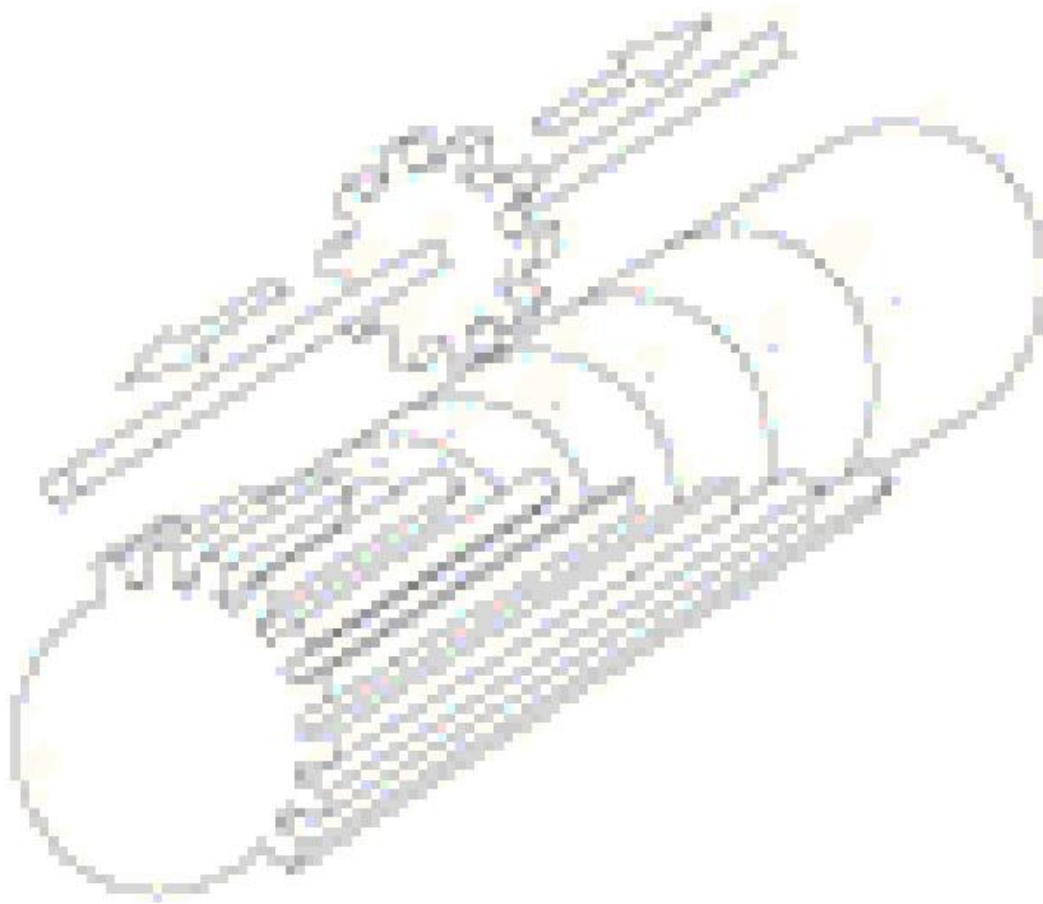




Leibniz

- Just a few years after Pascal, the German Gottfried Wilhelm Leibniz (co-inventor with Newton of calculus) managed to build a four-function (addition, subtraction, multiplication, and division) calculator that he called the ***stepped reckoner*** because, instead of gears, it employed fluted drums having ten flutes arranged around their circumference in a stair-step fashion. Although the stepped reckoner employed the decimal number system (each drum had 10 flutes), Leibniz was the first to advocate use of the binary number system which is fundamental to the operation of modern computers. Leibniz is considered one of the greatest of the philosophers but he died poor and alone.

Stepped Reckoner



Jacquard



- In 1801 the Frenchman Joseph Marie Jacquard invented a power loom that could base its weave (and hence the design on the fabric) upon a pattern automatically read from punched wooden cards, held together in a long row by rope. Descendents of these **punched cards** have been in use ever since (remember the "hanging chad" from the Florida presidential ballots of the year 2000?).

Jacquard's Loom

- By selecting particular cards for Jacquard's loom you defined the woven pattern



[Jacquard-card Making.]



Close up of a card





Jacquard's Loom



In a vivid demonstration of the power of his invention, Joseph-Marie Jacquard, using 10,000 punch cards, programmed a loom to weave a portrait of himself in black and white silk (above).

- Close up of a tapestry woven by the loom

Technology -vs- Jobs



- Jacquard's technology was a real boon to mill owners, but put many loom operators out of work. Angry mobs smashed Jacquard looms and once attacked Jacquard himself. History is full of examples of labor unrest following technological innovation yet most studies show that, overall, technology has actually increased the number of jobs.

Charles's Babbage



- By 1822 the English mathematician **Charles Babbage** was proposing a steam driven calculating machine the size of a room, which he called the ***Difference Engine***.



Difference Engine

- This machine would be able to compute tables of numbers, such as logarithm tables.
- He obtained government funding for this project due to the importance of numeric tables in ocean navigation.
- Construction of Babbage's Difference Engine proved exceedingly difficult and the project soon became the most expensive government funded project up to that point in English history.
- Ten years later the device was still nowhere near complete, acrimony abounded between all involved, and funding dried up. **The device was never finished.**



Babbage-Analytic Engine

- Babbage was not deterred, and by then was on to his next brainstorm, which he called the ***Analytic Engine***.
- This device, large as a house and powered by 6 steam engines,
- It was programmable, thanks to the punched card technology of Jacquard.
- Babbage saw that the pattern of holes in a punch card could be used to represent an abstract idea such as a problem statement or the raw data required for that problem's solution.



Babbage-Analytic Engine

- Babbage realized that punched paper could be employed as a storage mechanism, holding computed numbers for future reference.
- Because of the connection to the Jacquard loom, Babbage called the two main parts of his Analytic Engine the "Store" and the "Mill", as both terms are used in the weaving industry.
- The Store was where numbers were held and the Mill was where they were "woven" into new results.
- In a modern computer these same parts are called the ***memory unit*** and the ***central processing unit*** (CPU).



Babbage – Analytic Engine

- The Analytic Engine also had a key function that distinguishes computers from calculators: the conditional statement.
- A conditional statement allows a program to achieve different results each time it is run.
- Based on the conditional statement, the path of the program can be determined based upon a situation that is detected at the very moment the program is running.



Ada Byron

- Babbage befriended **Ada Byron**, the daughter of the famous poet Lord Byron
- Though she was only 19, she was fascinated by Babbage's ideas
- She began fashioning programs for the Analytic Engine, although still unbuilt.
- The Analytic Engine remained unbuilt (the British government refused to get involved with this one) but Ada earned her spot in history as the first computer programmer.
- Ada invented the subroutine and was the first to recognize the importance of looping.



US Census

- The next breakthrough occurred in America. The U.S. Constitution states that a census should be taken of all U.S. citizens every 10 years in order to determine the representation of the states in Congress.
- While the very first census of 1790 had only required 9 months, by 1880 the U.S. population had grown so much that the count for the 1880 census took 7.5 years. Automation was clearly needed for the next census.
- The census bureau offered a prize for an inventor to help with the 1890 census and this prize was won by Herman Hollerith,



Hollerith desk

- The ***Hollerith desk***, consisted of:
- a card reader which sensed the holes in the cards,
- a gear driven mechanism which could count (similar to Pascal's)
- A large wall of dial indicators to display the results of the count.

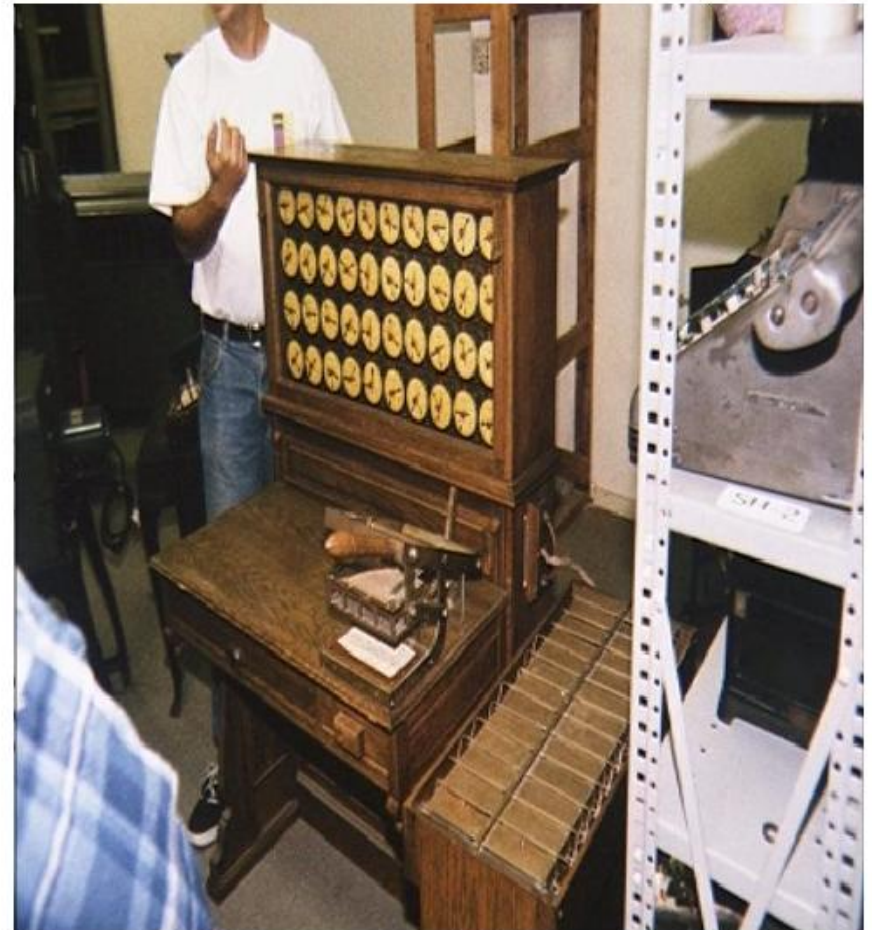
Hollerith Desk





Hollerith Desk

- Hollerith's technique was successful and the 1890 census was completed in only 3 years at a savings of 5 million dollars.



IBM



*sponsert
Frankfurt*

- Hollerith built a company, the Tabulating Machine Company which, after a few buyouts, eventually became International Business Machines, known today as **IBM**.

Hollerith's Innovation

- By using punch cards, Hollerith created a way to store and retrieve information.
- This was the first type of read and write technology



[illegible]



US Military

- The U.S. military desired a mechanical calculator more optimized for scientific computation.
- By World War II the U.S. had battleships that could lob shells weighing as much as a small car over distances up to 25 miles.
- Physicists could write the equations that described how atmospheric drag, wind, gravity, muzzle velocity, etc. would determine the trajectory of the shell, but solving such equations was extremely laborious.



US Military

- Human computers would compute results of these equations and publish them in ballistic "firing tables"
- During World War II the U.S. military scoured the country looking for (generally female) math majors to hire for the job of computing these tables, but not enough humans could be found to keep up with the need for new tables.
- Sometimes artillery pieces had to be delivered to the battlefield without the necessary firing tables and this meant they were close to useless because they couldn't be aimed properly.
- Faced with this situation, the U.S. military was willing to invest in even hair-brained schemes to automate this type of computation.

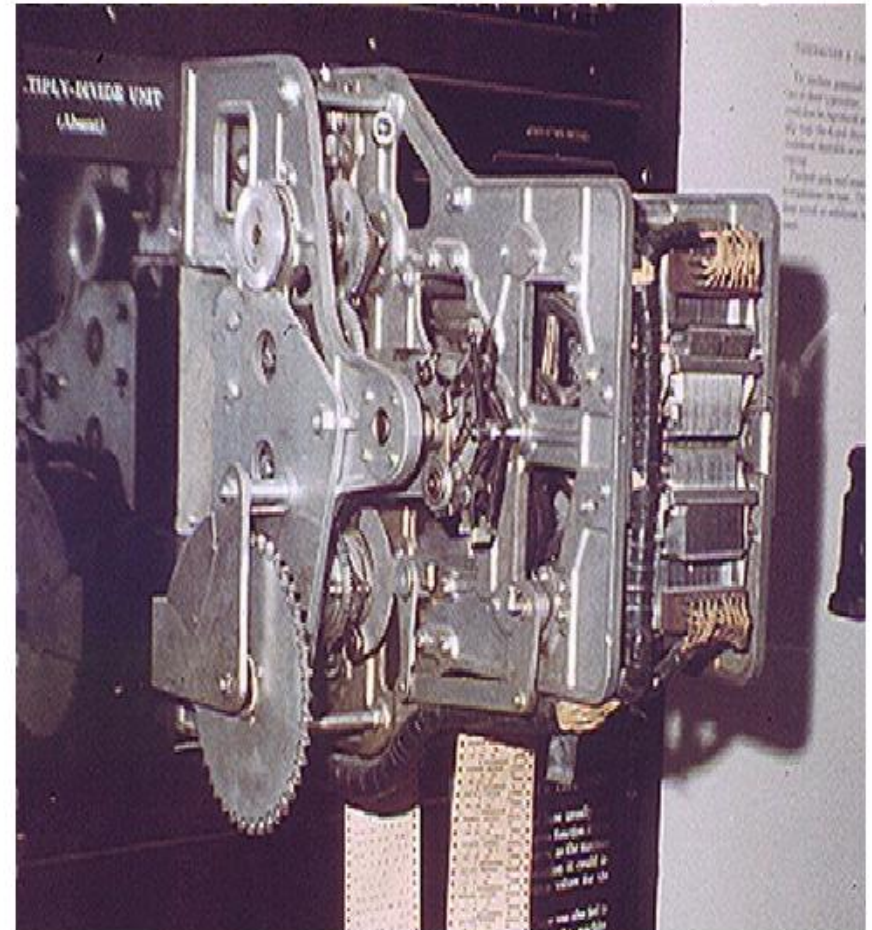
Mark I

- One early success was the Harvard **Mark I** computer which was built as a partnership between Harvard and IBM in 1944.
- This was the first programmable digital computer made in the U.S.
- But it was not a purely electronic computer. Instead the Mark I was constructed out of switches, relays, rotating shafts, and clutches.

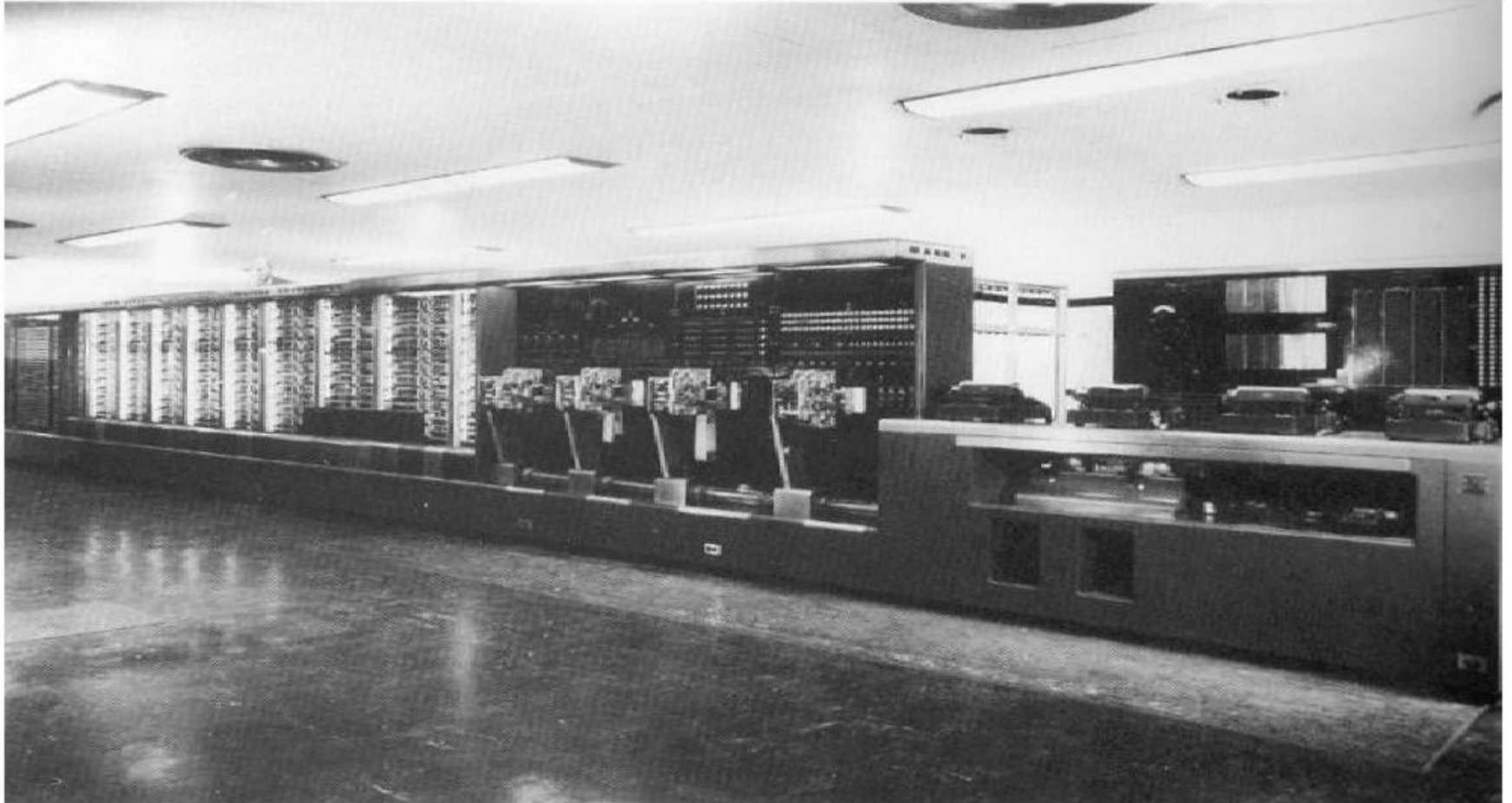


Mark I

- The machine weighed 5 tons, incorporated 500 miles of wire, was 8 feet tall and 51 feet long, and had a 50 ft rotating shaft running its length, turned by a 5 horsepower electric motor.
- The Mark I ran non-stop for 15 years, sounding like a roomful of ladies knitting.

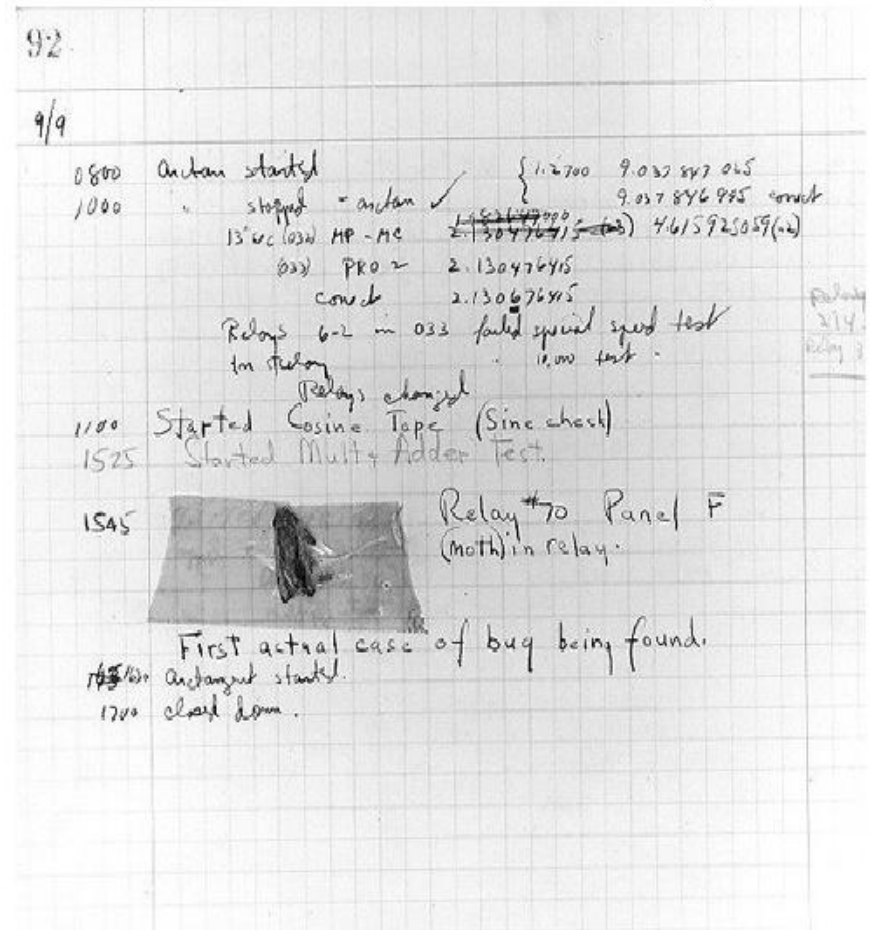


Mark I



The First Bug

- One of the primary programmers for the Mark I was a woman, **Grace Hopper**.
- Hopper found the first computer "bug": a dead moth that had gotten into the Mark I
- The word "bug" had been used to describe a defect since at least 1889 but Hopper is credited with coining the word "debugging" to describe the work to eliminate program faults.





Humor

- On a humorous note, the principal designer of the Mark I, **Howard Aiken** of Harvard, estimated in 1947 that six electronic digital computers would be sufficient to satisfy the computing needs of the entire United States.



The Future of Computers?

- IBM had commissioned this study to determine whether it should bother developing this new invention into one of its standard products (up until then computers were one-of-a-kind items built by special arrangement).
- Aiken's prediction wasn't actually so bad as there were very few institutions (principally, the government and military) that could afford the cost of what was called a computer in 1947.
- He just didn't foresee the micro-electronics revolution which would allow something like an **IBM Stretch** computer of 1959:



First Generation Computers

- The first electronic computer was designed at Iowa State between 1939-1942
- The Atanasoff-Berry Computer used the binary system(1's and 0's).
- Contained vacuum tubes and stored numbers for calculations by burning holes in paper

IBM Stretch - 1959



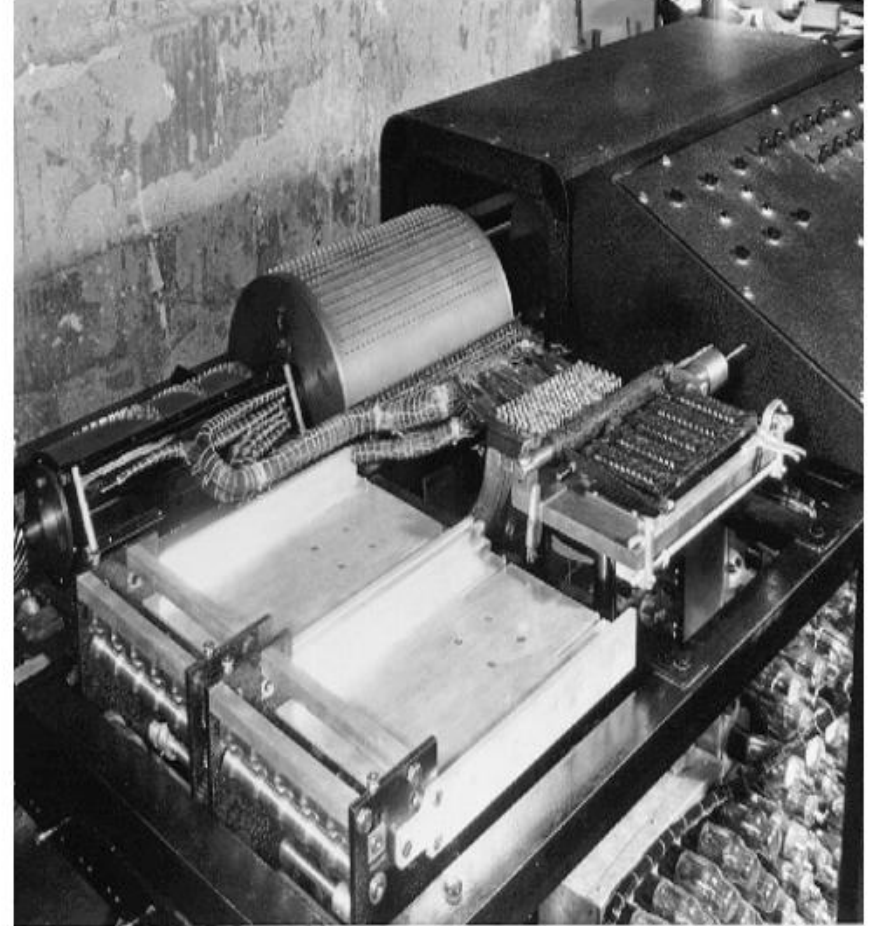
IBM Stretch - 1959



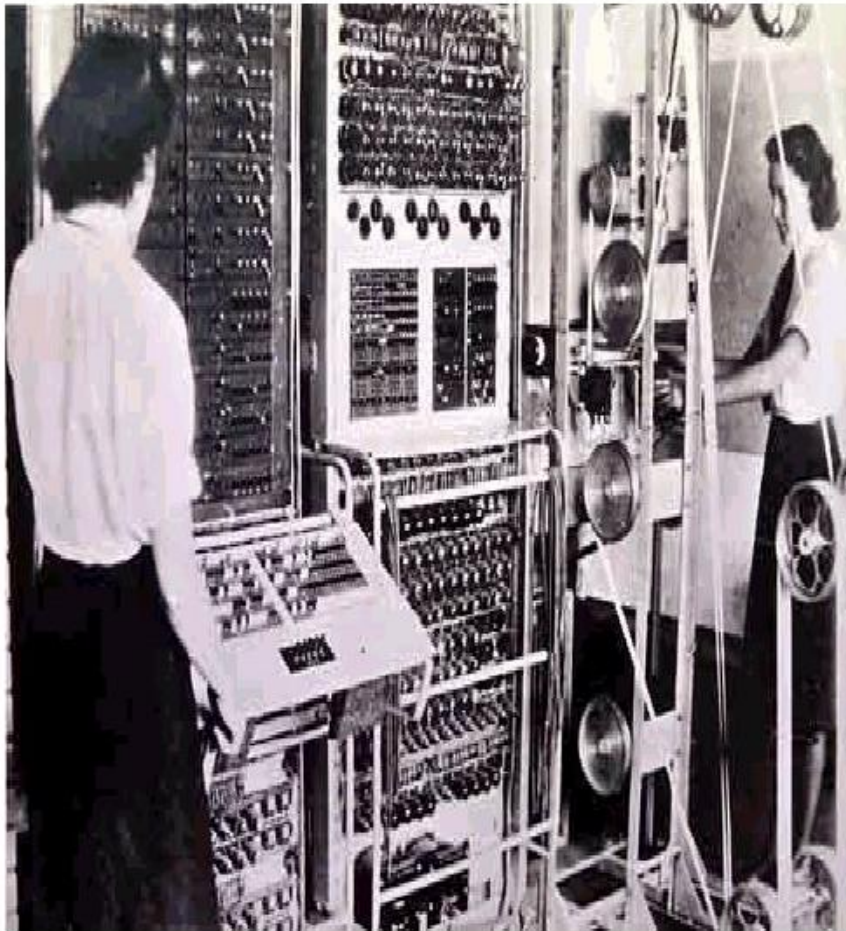


Atanasoff – Berry Computer

- One of the earliest attempts to build an all-electronic (that is, no gears, cams, belts, shafts, etc.) digital computer occurred in 1937 by **J. V. Atanasoff**,
- This machine was the first to store data as a charge on a capacitor, which is how today's computers store information in their main memory (**DRAM** or **dynamic RAM**). As far as its inventors were aware, it was also the first to employ binary arithmetic.



Colussus



- ***The Colossus***, built during World War II by Britain for the purpose of breaking the cryptographic codes used by Germany.
- Britain led the world in designing and building electronic machines dedicated to code breaking, and was routinely able to read coded Germany radio transmissions.
- Not a general purpose, reprogrammable machine.



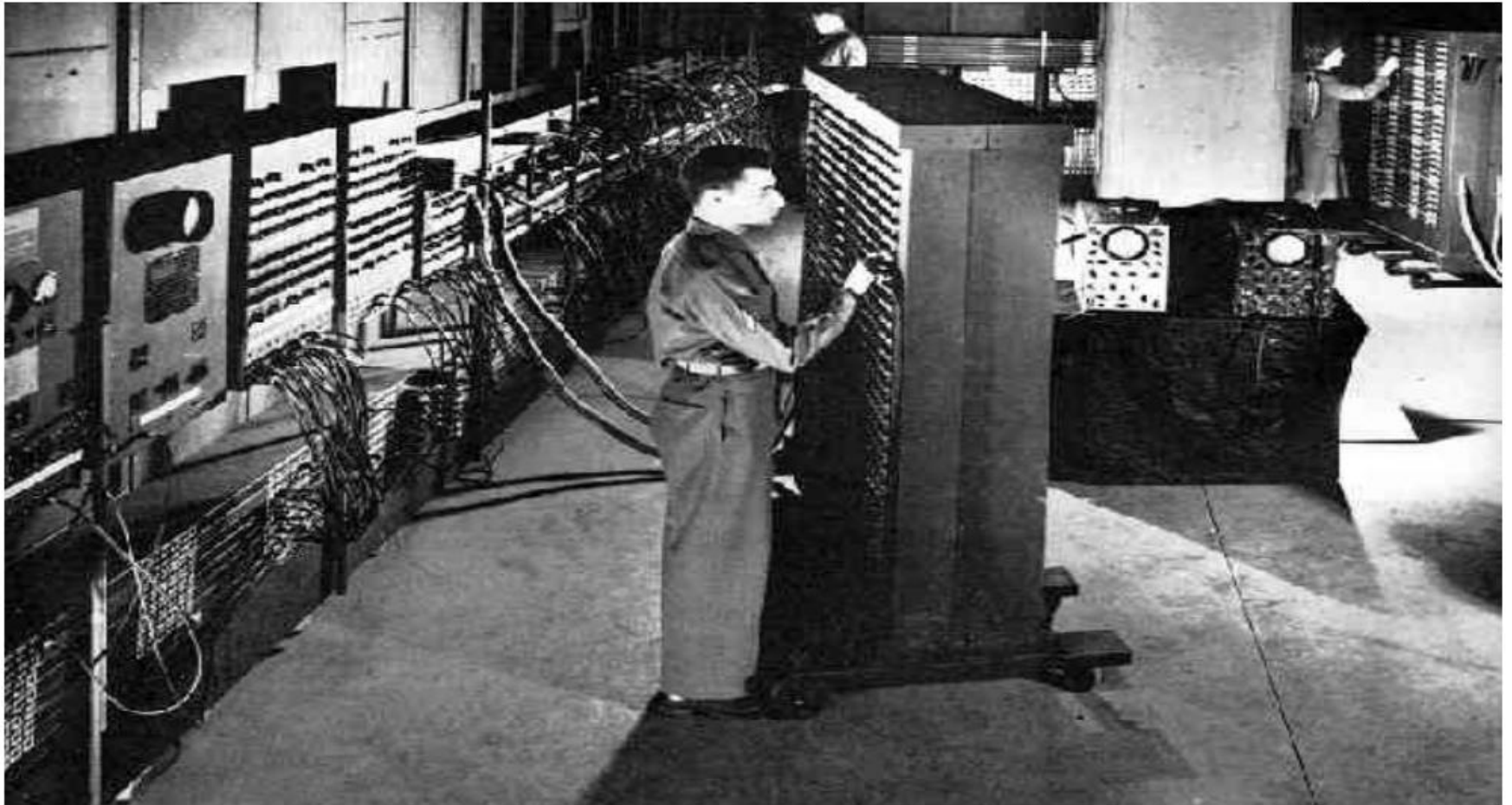
Eniac

- The title of forefather of today's all-electronic digital computers is usually awarded to **ENIAC**, which stood for Electronic Numerical Integrator and Calculator.
- ENIAC was built at the University of Pennsylvania between 1943 and 1945 by two professors, **John Mauchly** and the 24 year old **J. Presper Eckert**, who got funding from the war department after promising they could build a machine that would replace all the "computers"
- ENIAC filled a 20 by 40 foot room, weighed 30 tons, and used more than 18,000 vacuum tubes.

ENIAC



ENIAC

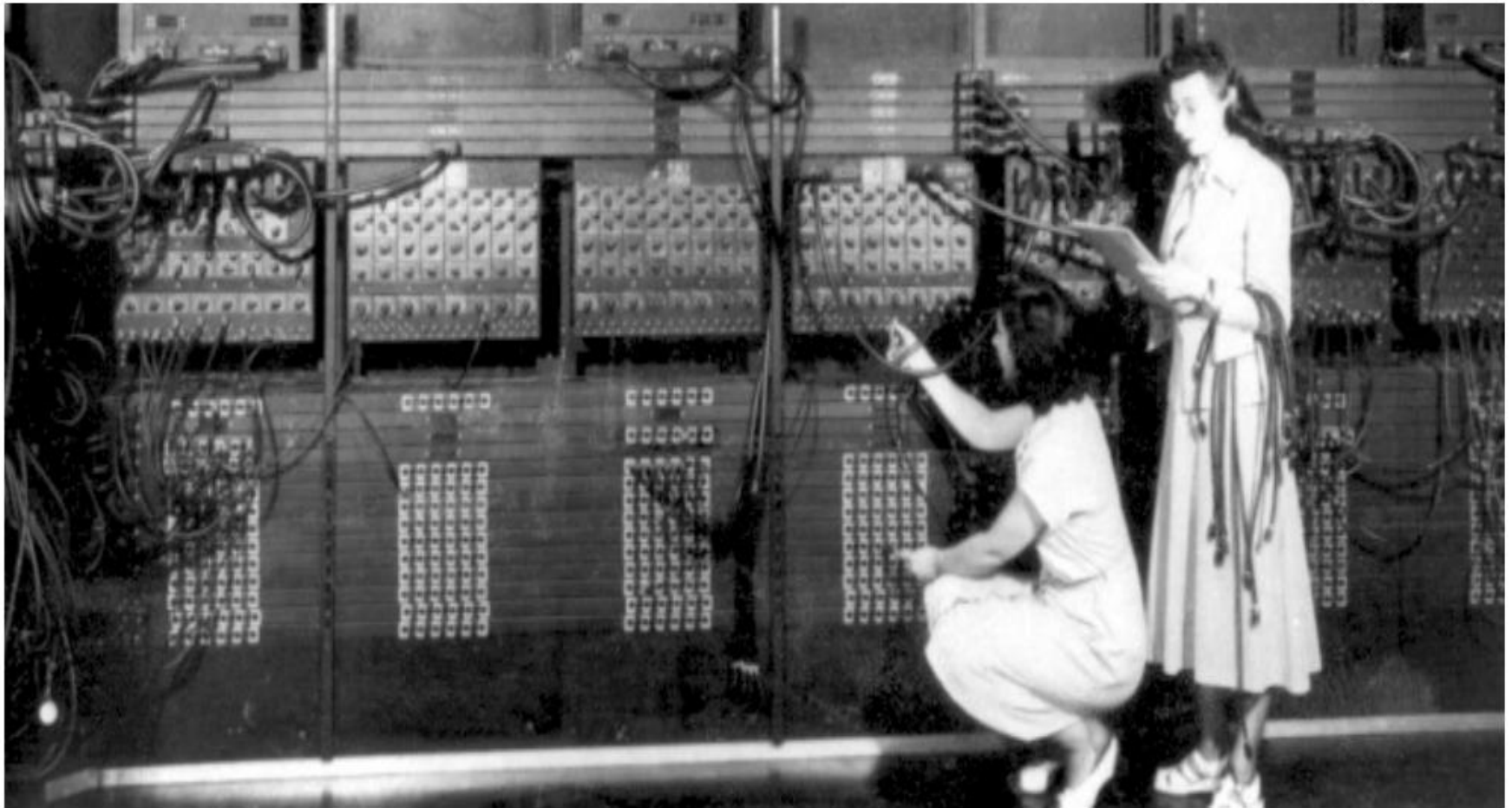




Programming the ENIAC

- To reprogram the ENIAC you had to rearrange the patch cords that you can observe on the left in the prior photo, and the settings of 3000 switches that you can observe on the right.
- To program a modern computer, you type out a program with statements like:
- $\text{Circumference} = 3.14 * \text{diameter}$
- To perform this computation on ENIAC you had to rearrange a large number of patch cords and then locate three particular knobs on that vast wall of knobs and set them to 3, 1, and 4.

Programming the ENIAC

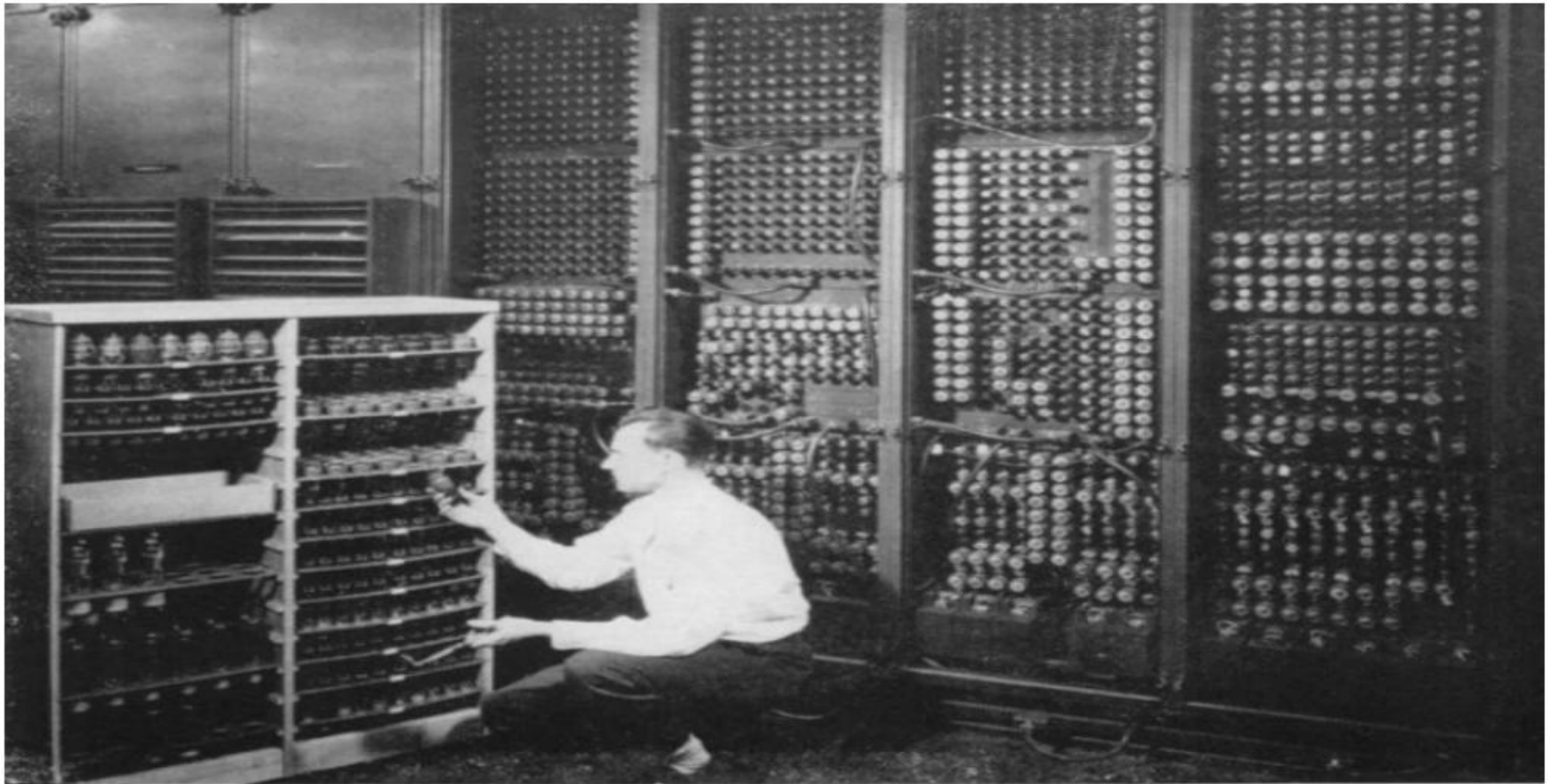




Problems with the ENIAC

- The ENIAC used 18,000 vacuum tubes to hold a charge
- Vacuum tubes were so notoriously unreliable that even twenty years later many neighborhood drug stores provided a "tube tester"

Replacing a vacuum tube



Replacing a bad tube meant checking among ENIAC's 19,000 possibilities.

The Stored Program Computer



- In 1945 John von Neumann presented his idea of a computer that would store computer instructions in a CPU
- The CPU(Central Processing Unit) consisted of elements that would control the computer electronically

The Stored Program Computer

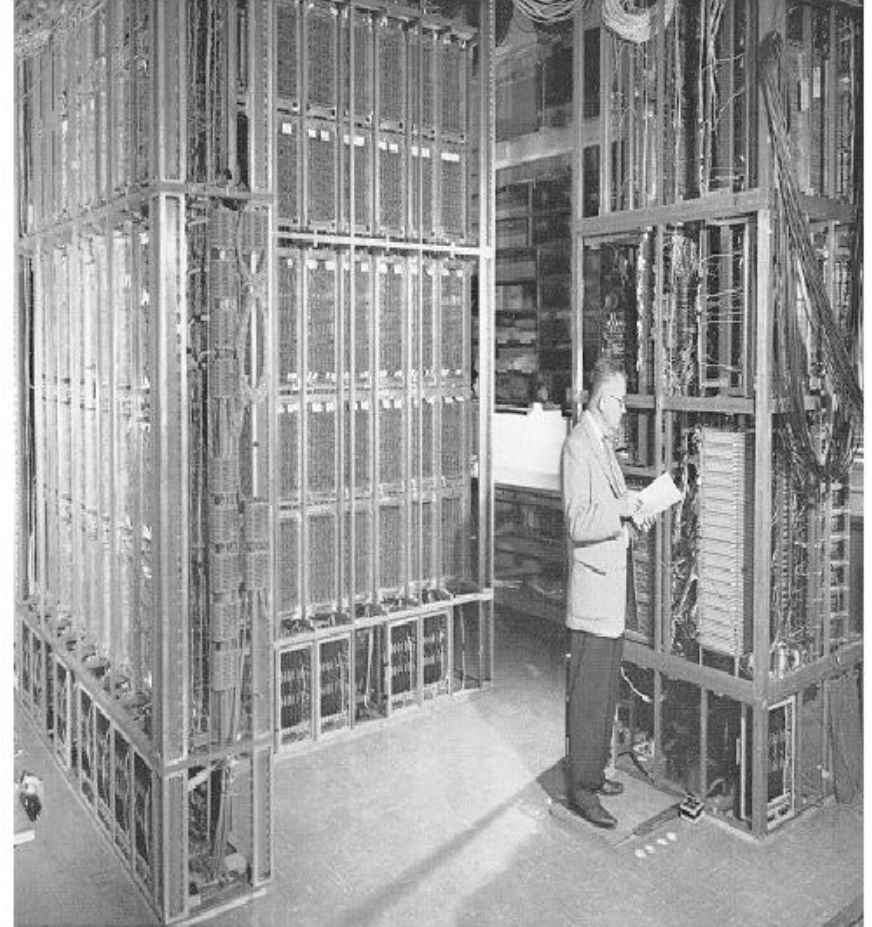


- The EDVAC, EDSAC and UNIVAC were the first computers to use the stored program concept
- They used vacuum tubes so they were too expensive and too large for households to own and afford



Edvac

- It took days to change ENIAC's program.
- Eckert and Mauchly's next teamed up with the mathematician **John von Neumann** to design **EDVAC**, which pioneered the **stored program**.
- After **ENIAC** and **EDVAC** came other computers with humorous names such as **ILLIAC**, **JOHNNIAC**, and, of course, **MANIAC**



Second Generation Computers



- In 1947, the transistor was invented
- The transistor made computers smaller, less expensive and increased calculating speeds.

Second Generation Computers



- Second generation computers also saw a new way data was stored
- Punch cards were replaced with magnetic tapes and reel to reel machines

Univac



- The **UNIVAC** computer was the first commercial (mass produced) computer.
- In the 50's, UNIVAC (a contraction of "Universal Automatic Computer") was the household word for "computer" just as "Kleenex" is for "tissue".
- UNIVAC was also the first computer to employ magnetic tape.





Third Generation Computers

- Transistors were replaced by integrated circuits(IC)
- One IC could replace hundreds of transistors
- This made computers even smaller and faster.



Fourth Generation Computers



- In 1970 the Intel Corporation invented the Microprocessor: an entire CPU on one chip
- This led to microcomputers- computers on a desk

Computer Programming in the '70's



- If you learned computer programming in the 1970's, you dealt with what today are called ***mainframe computers***, such as the IBM 7090 (shown below), IBM 360, or IBM 370.



Time-Sharing



- There were 2 ways to interact with a mainframe.
- The first was called ***time sharing*** because the computer gave each user a tiny sliver of time in a round-robin fashion.
- Perhaps 100 users would be simultaneously logged on, each typing on a ***teletype*** such as the following:



Teletype

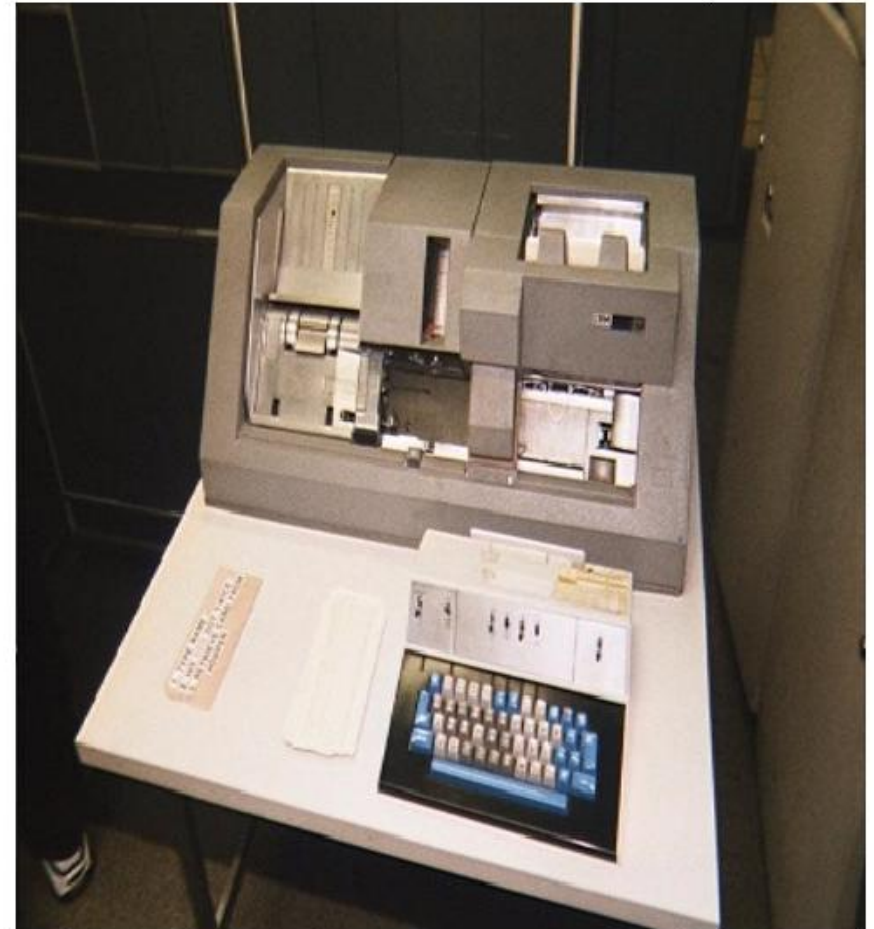
- A teletype was a motorized typewriter that could transmit your keystrokes to the mainframe and then print the computer's response on its roll of paper.
- You typed a single line of text, hit the carriage return button, and waited for the teletype to begin noisily printing the computer's response





Batch-Mode Processing

- The alternative to time sharing was **batch mode processing**, where the computer gives its full attention to your program.
- In exchange for getting the computer's full attention at run-time, you had to agree to prepare your program off-line on a **key punch machine** which generated punch cards.





Punch Cards

- University students in the 1970's bought blank cards a linear foot at a time from the university bookstore.
- Each card could hold only 1 program statement.
- To submit your program to the mainframe, you placed your stack of cards in the hopper of a card reader.
- Your program would be run whenever the computer made it that far.
- You often submitted your deck and then went to dinner or to bed and came back later hoping to see a successful printout showing your results



Programming Today

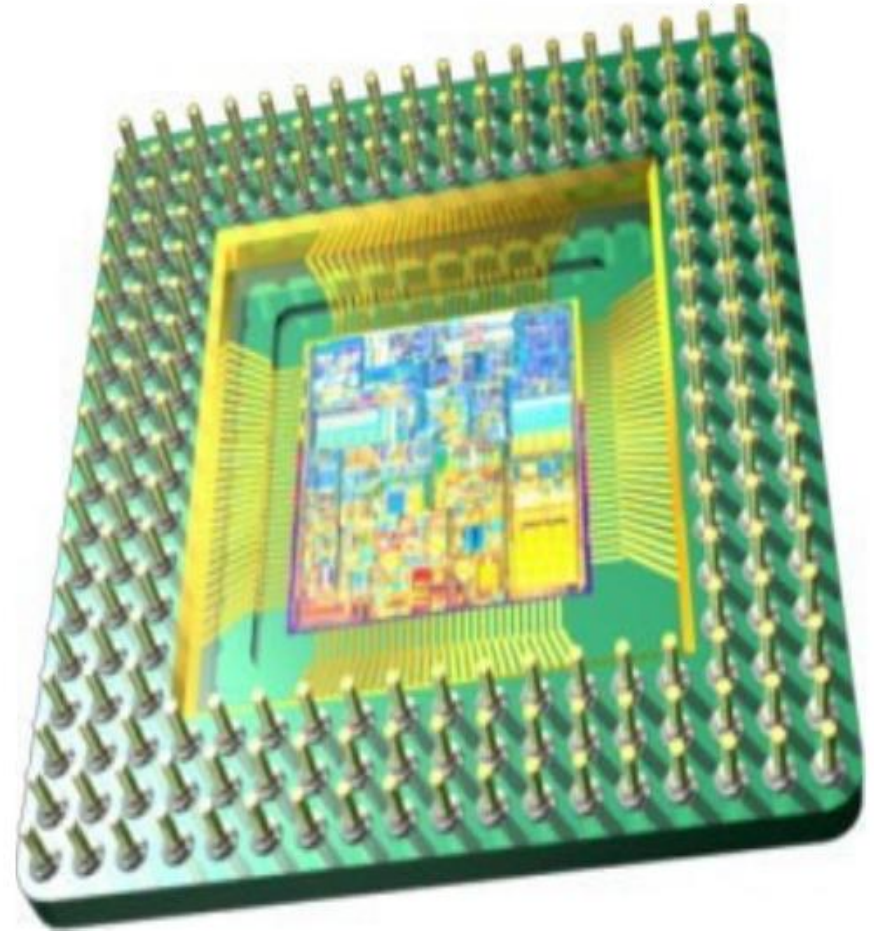
- But things changed fast. By the 1990's a university student would typically own his own computer and have exclusive use of it in his dorm room.



Microprocessor



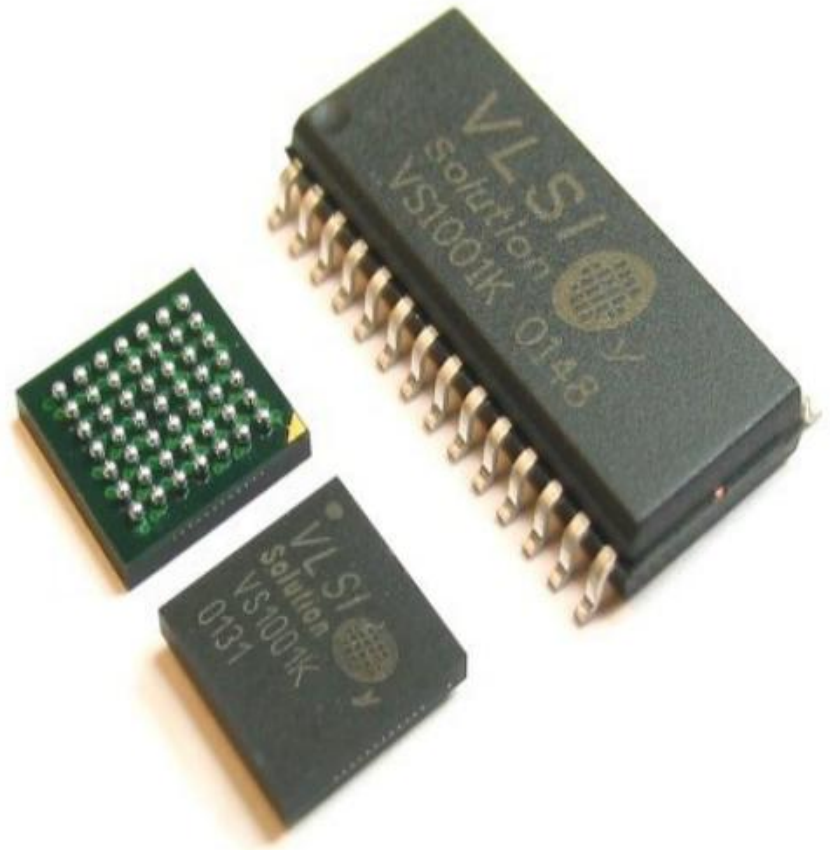
- This transformation was a result of the invention of the **microprocessor**.
- A microprocessor (uP) is a computer that is fabricated on an integrated circuit (IC).
- Computers had been around for 20 years before the first microprocessor was developed at **Intel** in 1971.





Microprocessor

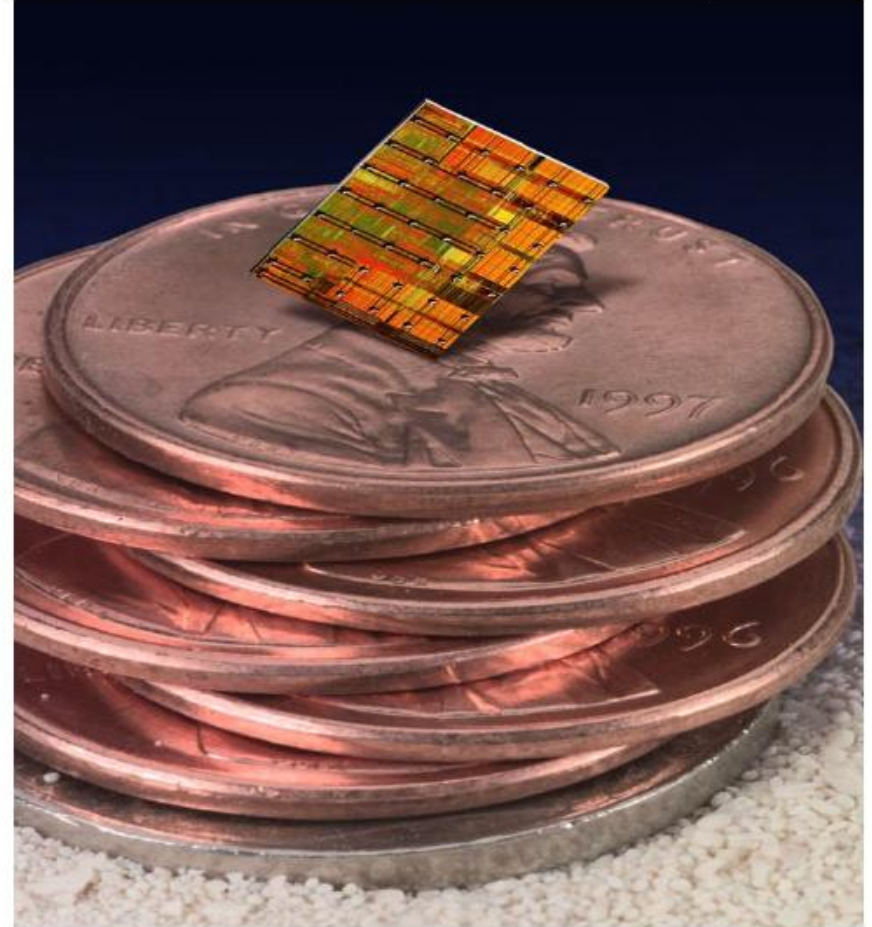
- The micro in the name microprocessor refers to the physical size.
- Intel didn't invent the electronic computer, but they were the first to succeed in cramming an entire computer on a single **chip** (IC)





Integrated Circuits

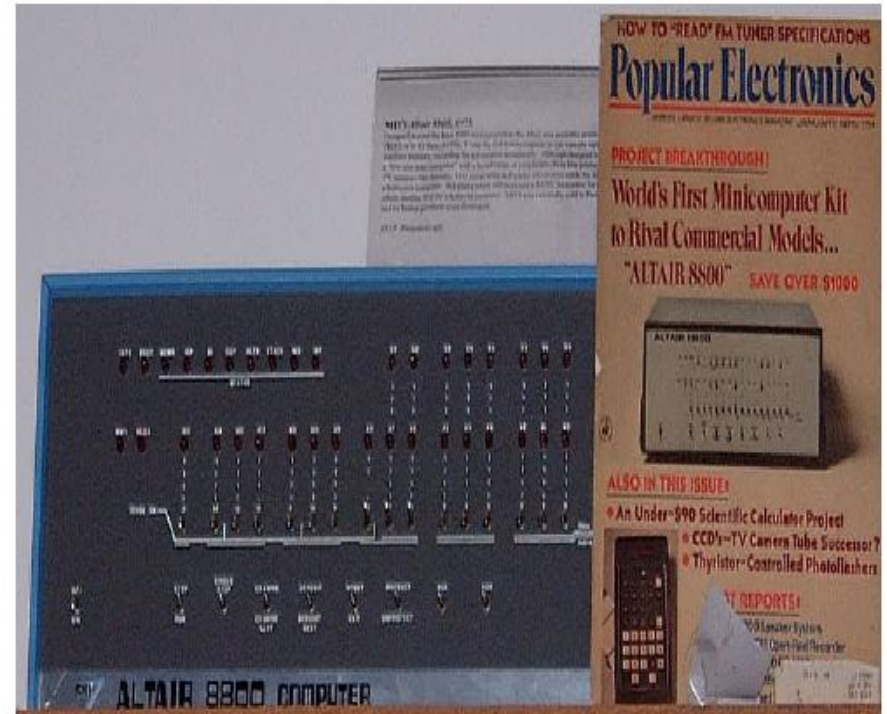
- The ***microelectronics revolution*** is what allowed the amount of hand-crafted wiring seen in the prior photo to be mass-produced as an ***integrated circuit*** which is a small sliver of silicon the size of your thumbnail



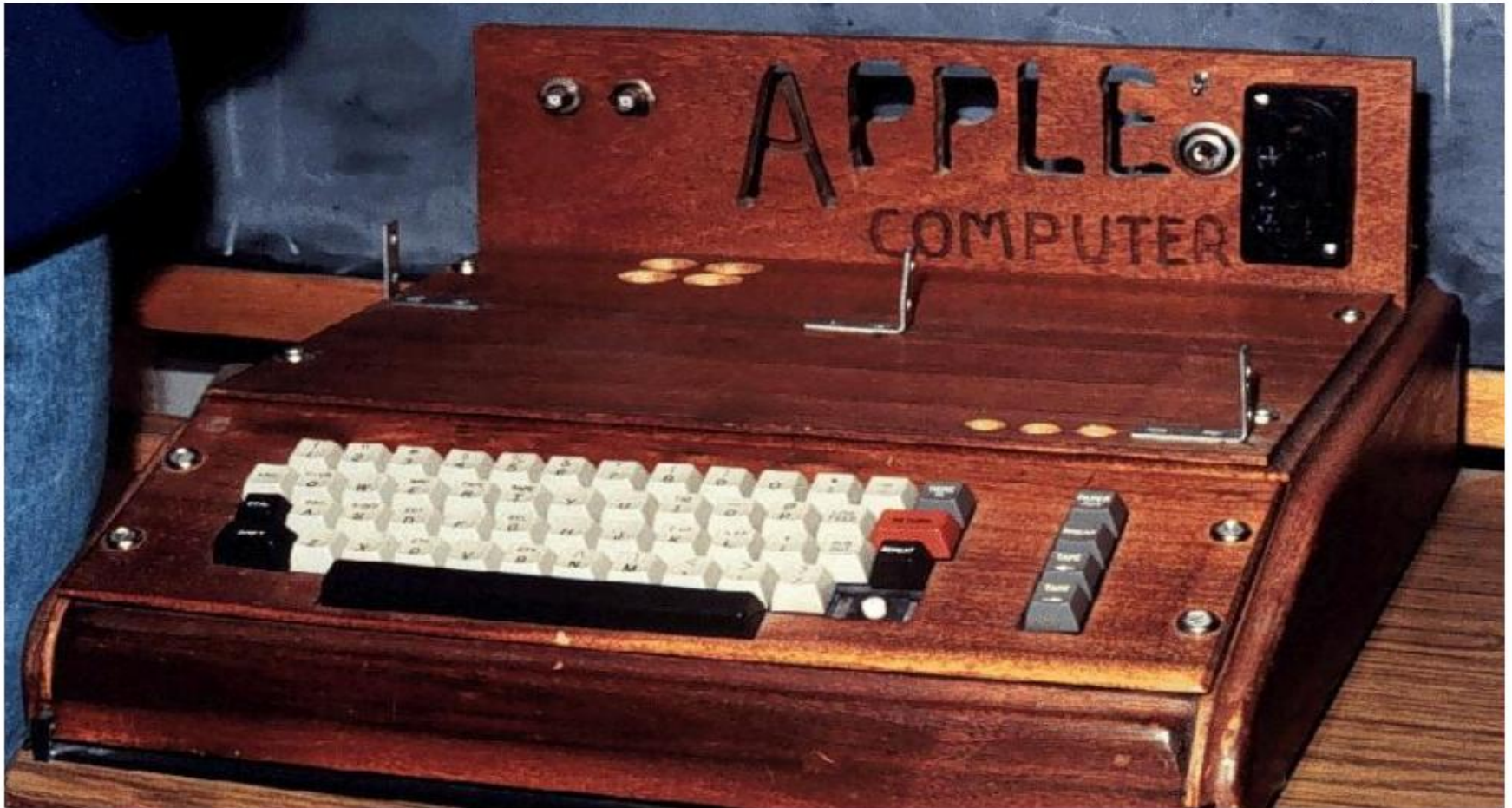


Integrated Circuits

- Integrated circuits and microprocessors allowed computers to be faster
- This led to a new age of computers
- The first home-brew computers is called the ALTAIR 8800



Apple 1 Computer - 1976



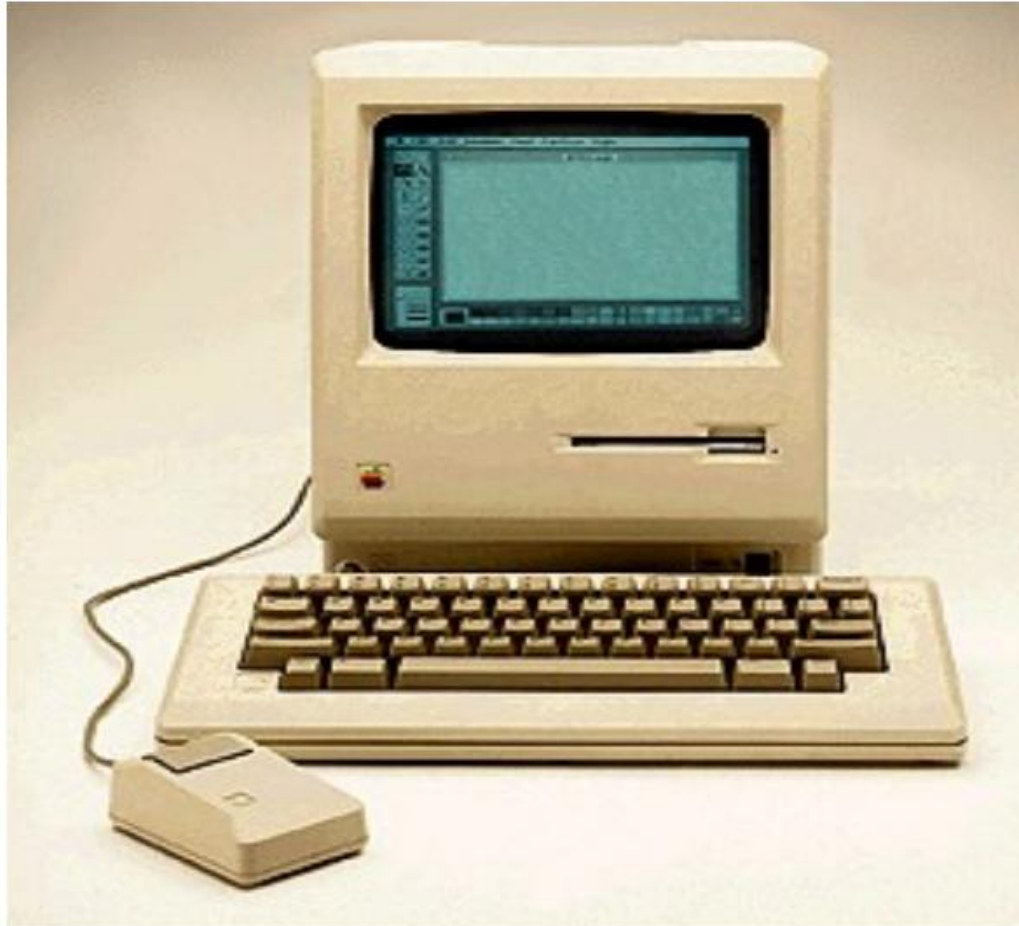
The IBM PC



Commodore 64



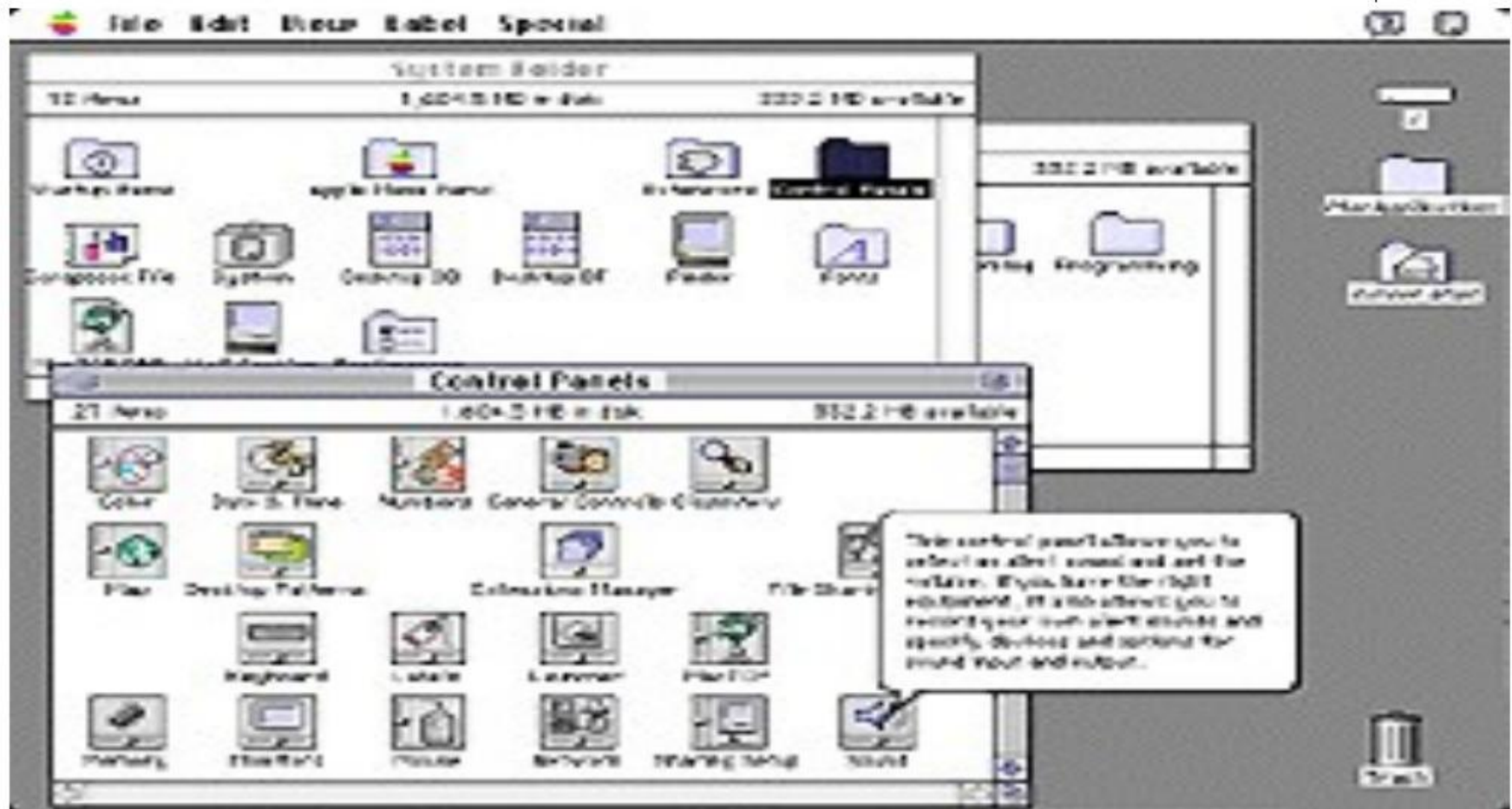
Apple Macintosh



The Amiga



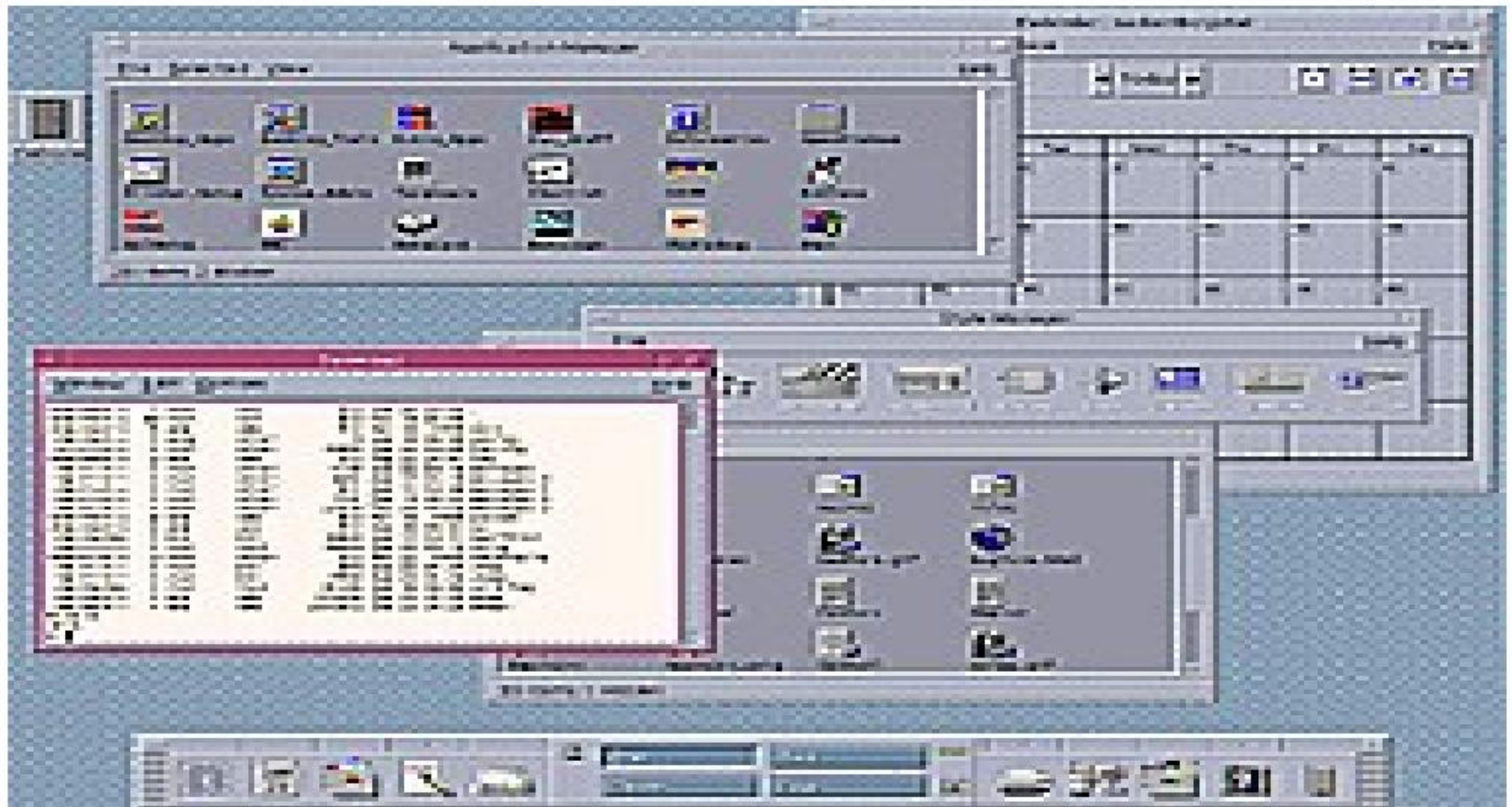
Macintosh System 7



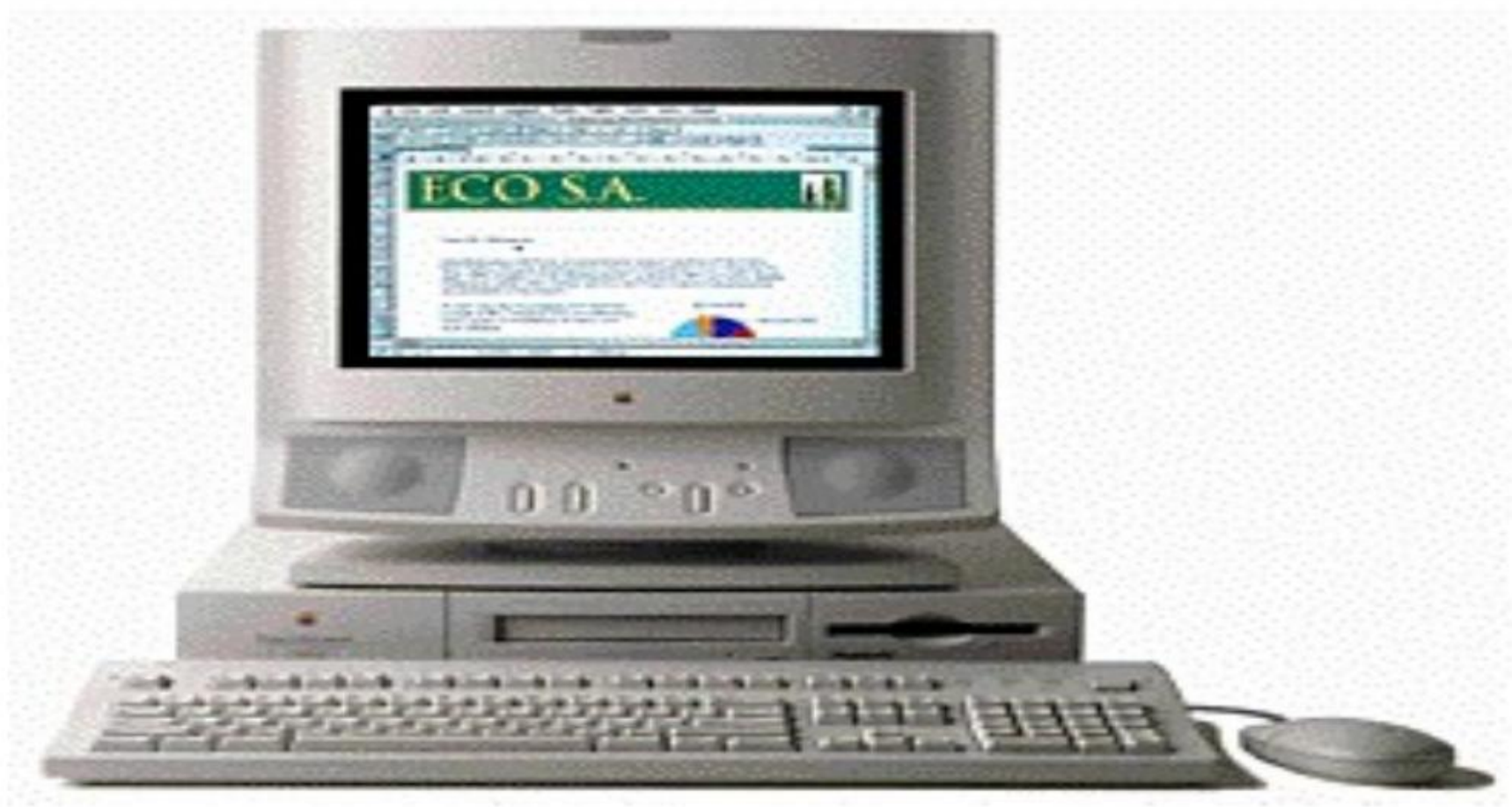
Apple Newton



Standard UNIX



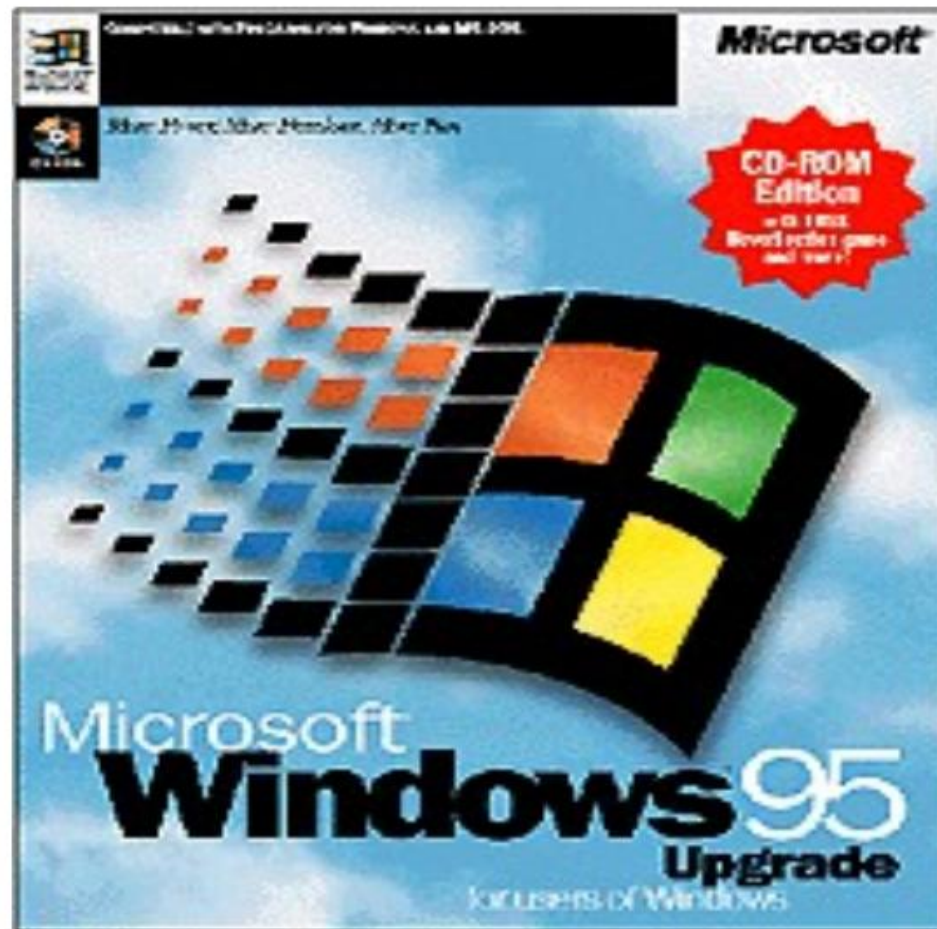
PowerPC



IBM OS/2



Windows 95





References

- Most of the information for this powerpoint was obtained from the following web page:
- <http://www.computersciencelab.com/ComputerHistory/History.htm>